Mechanistic Analysis of Over-Squashing and Attention Dilution in Hybrid GNNs

How far should a graph model listen? Large social graphs, two knobs, first results

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Collin McDevitt, Dhawal Dixit, Andy Jiang

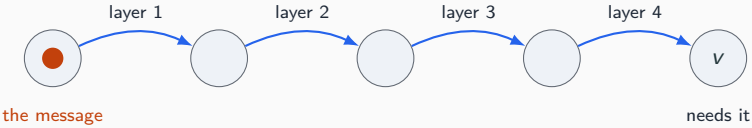
Supervisor: Dr. Greg Green MS Applied Data Science capstone

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Outline

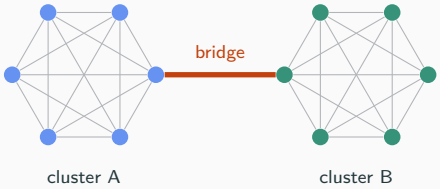
1. The question
2. Why social graphs
3. Measures
4. The knob problem
5. Pipeline
6. Results so far
7. Running now
8. Roadmap

Message passing: a game of telephone



- A graph neural network (GNN) updates each node from its **neighbours**
- One layer = one hop. With L layers a node hears at most L hops away
- Far information is **relayed** node by node, and mixed with everything else on the way

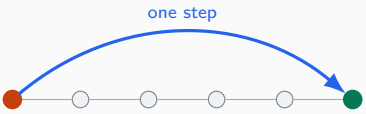
Over-squashing: many voices through one bridge



- Everything B knows must reach A through **one edge**
- Each node stores a fixed-size vector, so the messages get **compressed and lost**
- This is over-squashing (Alon & Yahav 2021; Topping et al. 2022)
- A narrow bridge = high **effective resistance** between A and B

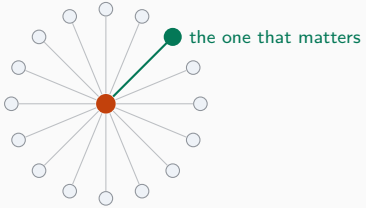
Attention: a direct line, but too many voices

Direct line



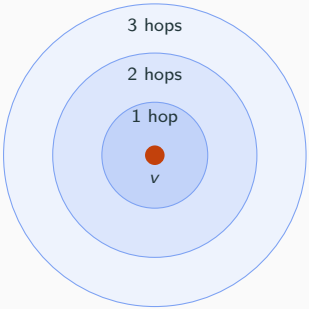
- Graph transformers let every node **look at every node**
- No relay, no bridge to squeeze through

Dilution



- Attention is split among **all** nodes; most are irrelevant
- The useful signal drowns: attention dilution

Task range: how far must a node listen?



- **Short range**: friends are enough
- **Long range**: needs friends of friends of friends
- Measured from the trained model (Bamberger et al., ICML 2025):

$$\rho_v = \frac{\sum_u I_{vu}\, d(v,u)}{\sum_u I_{vu}}$$

I_{vu}: how much u's input moves v's output; d: hops

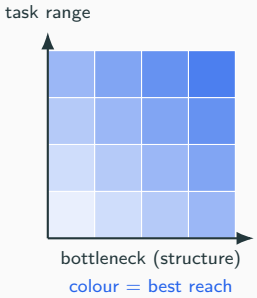
- Average distance a node "listens" to

Research question

Question

Do **graph structure** (bottlenecks) and **task range** predict how far a model's attention should reach?

- Too short a reach ⇒ over-squashing
- Too long a reach ⇒ dilution
- Measures explain *why*: Jacobian, attention entropy, range



Deliverable: a **phase diagram** (sketch, no data)

Why social graphs?

- **Small world**: most users are 3–6 hops apart
- **Hubs**: a few accounts have very many links
- So a few hops already reach huge crowds: both failure modes are likely
- Real tasks: content labels, user activity

A gap in the benchmarks

- **LRGB**: vision, chemistry and quantum-chemistry graphs (Dwivedi et al. 2022)
- **ECHO**: synthetic long-diameter graphs and molecules (Miglior et al., ICLR 2026)
- **No social graphs** in either

Older Peptides, river and tree experiments: appendix.

Datasets (SNAP) and real targets

Dataset	Graph	Target	Mean deg.	Eff. diam.
twitch-gamers	168,114 users, 6.8M mutual follows	`mature` (explicit content, binary)	11.6	5.8
reddit-body	35,776 subreddits, 124,330 links	`LIWC_Money` (regression)	4.3	5.7
twitch-de	9,498 users, 153,138 edges	`mature`	22.9	3.0
higgs-follow	35,593 users, 528,532 follows	`active_after`: active after the Higgs news (4 Jul 2012)	16.7	3.7

- `activate_passive`: same Higgs label, scored only on the $\sim$11% of users with **no earlier activity**, so any signal must come from neighbours
- reddit target: neighbours add the most (ridge R^2 0.108 own $\rightarrow$ 0.185 with 4 hops)

Degree, diameter: means over test subgraphs. Higgs retweet / reply layers before the cutoff: too sparse (mean degree 1.4 / 0.16).

Effective resistance collapses to degree

$$R(u,v) = (e_u - e_v)^\top L^+ (e_u - e_v)$$

- Low R: many wide paths. High R: a bottleneck
- On large, well-connected graphs (von Luxburg, Radl & Hein 2014):

$$R(u,v) \;\to\; \tfrac{1}{d_u} + \tfrac{1}{d_v}$$

- So R mostly measures degree, not bottlenecks
- We use **excess resistance**

$$R_{\text{ex}}(u,v) = R(u,v) - \left(\tfrac{1}{d_u} + \tfrac{1}{d_v}\right)$$

Graph	$\rho_S\left(R,\ \frac{1}{d_u}+\frac{1}{d_v}\right)$	mean R_{ex}
twitch-gamers	0.90	0.39
reddit	0.77	0.49
twitch-de	0.998	0.02
higgs-follow	0.98	0.09
higgs-retweet	0.07	–

ρ_S: Spearman correlation, on test subgraphs. Only the very sparse retweet layer escapes the collapse.

Three model measures

Jacobian sensitivity

$$J(u, v) = \left\| \frac{\partial h_v^{(L)}}{\partial h_u^{(0)}} \right\|_F$$

- Does v's output move when u's input moves?
- Over-squashing: J **decays** with hops and resistance

Attention entropy

$$\tilde{H} = \frac{-\sum_j a_{ij} \log a_{ij}}{\log N} \in [0, 1]$$

- $\tilde{H} \approx 1$: attention spread evenly over everyone
- High entropy = **dilution**
- Exact for softmax; implicit rows for Performer

Range

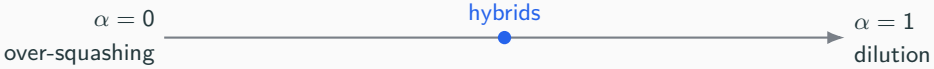
$$\rho_v = \frac{\sum_u I_{vu}\, d(v, u)}{\sum_u I_{vu}}$$

- How far the trained model listens
- Node level, from the Jacobian (Bamberger et al. 2025)
- Distance in hops or (excess) resistance

The hybrid model and its α knob

$$x' = x + (1-\alpha)\,\mathrm{Norm}\big(\mathrm{MPNN}(x)\big) + \alpha\,\mathrm{Norm}\big(\mathrm{Attn}(x)\big)$$

- $\alpha = 0$: pure message passing $\alpha = 1$: pure attention (GraphGPS-like, Rampášek et al. 2022)
- Plan: sweep α and find the best mix per graph and task



But: as built, training could undo α. Next slide.

Why α did nothing: the norms absorb it

BatchNorm has a learnable scale γ: $\mathrm{Norm}(h) = \gamma\,\hat{h} + \beta$

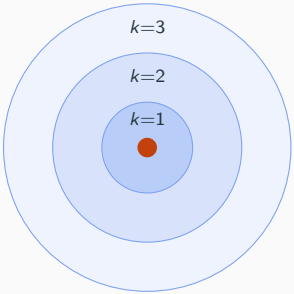
$$(1-\alpha)\,\mathrm{Norm}(h_{\text{loc}}) + \alpha\,\mathrm{Norm}(h_{\text{glb}}) = \underbrace{(1-\alpha)\gamma_{\text{loc}}}_{\text{learned}}\,\hat{h}_{\text{loc}} + \underbrace{\alpha\,\gamma_{\text{glb}}}_{\text{learned}}\,\hat{h}_{\text{glb}} + \dots$$

- Training can rescale either branch freely
- Any α strictly between 0 and 1 gives the **same** model family
- α only changed the starting point, not the model

Fix 1: affine-free α

Remove the learnable scale from the branch norms (`branch_affine=false`). Now α is a real, fixed weight.

Fix 2: the attention radius k

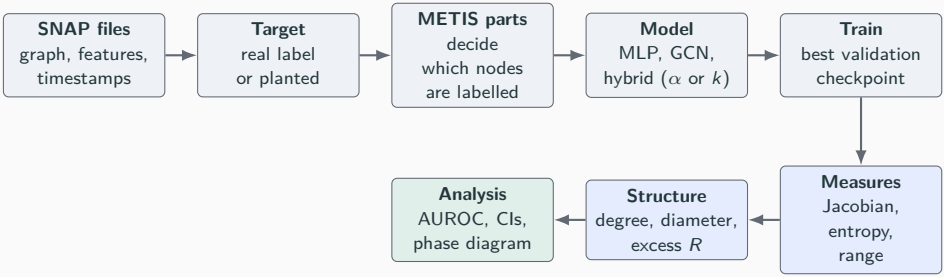


$k = \infty$: the whole graph

- Attention may only look at nodes **within k hops**
- $k = 0$: pure message passing; $k = \infty$: full attention
- Larger k: shortcuts past bottlenecks (less over-squashing)
- . . . but more nodes per softmax (more dilution)
- The best k *is* the trade-off we study
- Learned weights **cannot undo** it

Related architectures: DRew (Gutteridge et al. 2023, k-hop rewiring), Shortest Path Networks (Abboud et al. 2022), Graphormer's distance bias (Ying et al. 2021). Our contribution: the radius as a *diagnostic* knob, read with the measures.

From a SNAP file to a measured model



- **Subgraph mode** (so far): train and test inside each part
- **Full-graph mode** (now): parts only pick the labelled nodes; the model sees the **whole graph**
- **No leakage** on Higgs: inputs are follow degree and activity counts *before* the cutoff only

Models and baselines

Baselines

- **MLP**: own features only, no graph
- **GCN**: plain message passing

Hybrid

- Local branch: GCN-normalised message passing (GINE in the first sweep; its sum was unstable on hubs)
- Global branch: attention
- Branch norms without learnable scale

Three kinds of attention

- **Exact softmax**: small graphs
- **Performer** (FAVOR+): linear cost, for $k = \infty$ on big graphs (Choromanski et al. 2021)
- **Radius-masked softmax**: only pairs within k hops

Metric: test AUROC (binary targets), mean $\pm$ std over split and model seeds

Colab notebooks

Notebook	What it does	Status
`00_quickstart`	one short run, end-to-end check	**done**
`01_fetch_datasets`	download, parse, structure stats (S0)	**done**
`02_s2_<dataset>`	first α sweep, one notebook per dataset	**done** (Higgs)
`03_s2_results`	collect runs into tables and α curves	built
`04_higgs_affine_alpha`	affine-free α sweep, 500-node subgraphs	**done**
`05_higgs_radius`	radius sweep, 500-node subgraphs	**done**
`06_higgs_full_graph`	full-graph α and radius sweeps	**running**
`07_higgs_planted`	planted long-range targets	**being built**

Each notebook mounts Drive, clones the repo and resumes: finished runs are skipped, interrupted runs continue from their checkpoint.

Higgs, first sweep: nothing beats the MLP

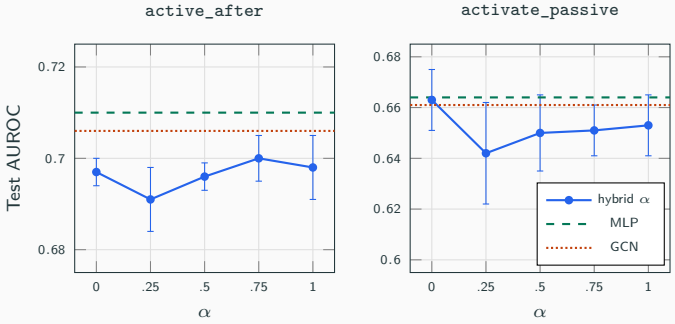
Model	Test AUROC
MLP (no graph)	**0.687** $\pm$ 0.000
GCN	0.684 $\pm$ 0.000
$\alpha = 1$	0.677 $\pm$ 0.003
$\alpha = 0.5$	0.669 $\pm$ 0.026
$\alpha = 0.75$	0.666 $\pm$ 0.019
$\alpha = 0$	0.652 $\pm$ 0.034
$\alpha = 0.25$	0.648 $\pm$ 0.003

higgs-follow, `active_after`, 2k-node subgraphs, 3 seeds

- Setup: GINE local branch, Performer attention, flawed α (learnable norm scales)
- The graph adds nothing over own features
- GINE's sum aggregation was **unstable on hubs**

$\Rightarrow$ Fix the knob (affine-free), switch to a GCN-normalised local branch

Higgs, affine-free α: still no gain over the MLP



Mean ± std, 3 split seeds × 2 model seeds. `active_after` $\alpha = 0, .25, .5$: 4 of 6 runs (Drive error).

- Notebook 04: 500-node subgraphs, exact softmax attention
- MLP 0.710 / 0.664; GCN 0.706 / 0.661
- Hybrids: 0.691–0.700 / 0.642–0.663
- No graph model beat the MLP, on either target
- α barely matters

Diagnosis: cutting the graph removed the signal

Part size	Edges kept	Users	Own only	+ neighbours (full)	+ neighbours (in part)
500	32%	all	0.672	**0.690**	0.673
500	32%	passive	0.677	**0.705**	0.690
2000	47%	all	0.694	**0.707**	0.700

Logistic regression on own features + 1- and 2-hop neighbour averages, split by METIS part; test AUROC.

- There *is* a neighbour signal: real but modest, larger for passive users
- Inside 500-node parts only 32% of edges survive, and most of the signal goes with them

Takeaway

On small-world social graphs, subgraph training throws away the edges the models need. So: **train on the full graph**.

Full-graph sweeps on Higgs running

Design

- Whole 35.6k-node graph for message passing and attention; parts only pick labelled nodes
- α **sweep**: $\alpha \in \{0, .25, .5, .75, 1\}$, affine-free, Performer
- **Radius sweep**: $k \in \{0, 1, \infty\}$ ($k = 2$: 299M pairs, does not fit)
- $+$ MLP, GCN; both targets; 3 split $\times$ 2 model seeds
- Done on 500-node subgraphs: $k = 1$ best by a hair, $k \geq 2$ about 0.005–0.01 lower, all within noise; no radius beats the MLP

What would support the thesis

- Graph models **beat the MLP** once the edges are back
- The gain is larger on `activate_passive`
- The best k is **between** 0 and ∞, and entropy rises with k

Results pending

Full-graph α and radius sweeps.

Planted targets: a known range on a real graph being built

- Keep the real graph and features; **plant** a signal z_u with known range k
- `shell_mean(k)`: predict the mean of z_u over nodes *exactly* k hops away (sanity)
- `retrieve(k)`: fetch the value of the node k hops away whose key matches your query (main test)
- Higgs, $k = 1$–3; MLP, GCN, MPNN, attention, hybrid, radius 1 ($k = 4$: 97% of graph in reach)

Caveat: small world (effective diameter ≈3.7, mean degree ≈30), so $k \geq 3$ is close to the whole graph.

What would support the thesis

- Measured range **tracks** planted k (validates the measure)
- The best radius **grows with** k
- MP fails `retrieve` at large k; full attention loses to dilution

Results pending

Planted-target sweep (S3).

Roadmap

Phase	What	Status
S0	Social data, subgraphs, structure stats (excess resistance)	**done**
S1	Performer attention + node-level models	**done**
S2	Sweeps on real targets (now on the full graph)	**running**
S3	Planted targets with known range k	**being built**
S4	Phase diagram: structure $\times$ task range $\rightarrow$ best setting	planned
S5	Graph foundation model (GFM) benchmark	planned
S6	Benchmark release and final writeup	planned

Cut order if time is short: GraphAny, then Higgs multilayer analysis, then packaging polish. S1–S4 are never cut.

Where it leads: phase diagram, GFMs, benchmark

S4 Phase diagram

- x: excess resistance
- y: task range (planted k)
- Colour: best α / radius
- Overlay: attention entropy

S5 Do GFMs reach far enough?

- Fixed-backbone GFMs **under-reach** when task range differs from training (Gutteridge, Bronstein & Dong 2026; MP only)
- Test **GraphPFN** (Eremeev et al., ICML 2026), maybe GraphAny, on planted tasks; measure their range
- NodeGround: tuned GNNs still beat GFMs; no long-range analysis

S6 Benchmark

- First long-range benchmark on **social** graphs
- Released as generator code + node indices + seeds
- Rebuilt from public SNAP files

Takeaways

- GNNs pass messages hop by hop: far information is **squashed**; attention reaches far but **dilutes**
- Social graphs are small-world with hubs, and no long-range benchmark covers them
- Resistance collapses to degree there, so we use **excess resistance**
- The α knob could be undone by the norms: fixed (affine-free) and joined by the **attention radius** k
- On Higgs, no graph model beat the MLP yet: subgraph cuts removed most of the neighbour signal
- Now: full-graph sweeps and planted targets with known range; then the phase diagram, GFMs and a benchmark

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Appendix: background experiments

- Earlier work, before the move to social graphs. Kept as background
- **Peptides-func** (LRGB molecules): pilot replication, rebuilt-hybrid calibration (G0)
- **Reddit pilot**: virtual node vs SAGE by hop band
- **Untrained demos**: the measures behave as theory predicts
- **Synthetic trees**: message passing fails a 4-hop query
- **LamaH-CE rivers**: does upstream growth predict the hybrid's gain?

These used the old α-blend (with learnable norm scales), so their α curves inherit the knob problem.

Peptides-func: data and the old plan

- 15,535 molecules; ≈151 nodes, diameter ≈57 (Dwivedi et al. 2022)
- 10-label graph classification, macro AP
- RWSE positional encoding (20 steps)
- Near-trees, so $R \approx$ hops: "resistance beats hops" cannot be shown here
- Bamberger et al. find it short-range

Gate G0

- Rebuilt 4-layer hybrid, $\alpha = 0.5$, seed 0
- Must reach test AP ≥ 0.652 (the pilot GPS value)
- Result: 0.6259, not passed
- No further Peptides GPU work

Pilot LRGB replication on Peptides-func [pilot]

Model	Val AP	Test AP	Published
GPS	0.6702	**0.6518**	0.6535
GCN	0.6545	**0.6263**	≈0.686

- RWSE, 200 epochs (patience 40), 1 seed
- GPS reproduces the published value
- GCN is 0.06 short of Tönshoff et al.'s tuned value: a partial replication
- G0's 0.652 threshold comes from this GPS run

Pilot mechanistic notebook [pilot]

Jacobian vs resistance (log-log slope, 240 pairs)

	GINE	Transf.	GPS
slope	-1.76	-0.48	-0.81
r	-0.78	-0.41	-0.69
test AP	0.54	0.27	0.24

Entropy vs size (400 graphs)

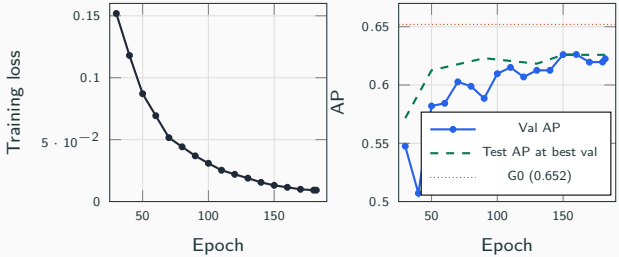
- Raw H rises for both models ($r = +0.88$, $+0.91$)
- $H/\log N$ rises *only for GPS* ($r = +0.60$), not for the transformer ($r = -0.09$)

Caveats: not citable as results

- 30 epochs, 1 seed; the attention models did not train
- Atom codes passed through `nn.Linear`; no positional encoding
- Entropy computed on head-averaged attention

Nine bugs were fixed afterwards in `gnn_mech`

Gate G0 calibration on Colab: below threshold



`pilot_equivalence.yaml`: 4 layers, $\alpha = 0.5$, seed 0, 404,654 params, T4, $\approx$19 s/epoch

- Best val AP 0.6281 at epoch 142 (stopped at 182); **test AP 0.6259**, 0.026 short of 0.652
- Loss keeps falling to 0.009 while val AP plateaus near 0.62 from epoch 90: overfitting
- Measurements not saved: the Hessian graph range ran out of memory on the T4

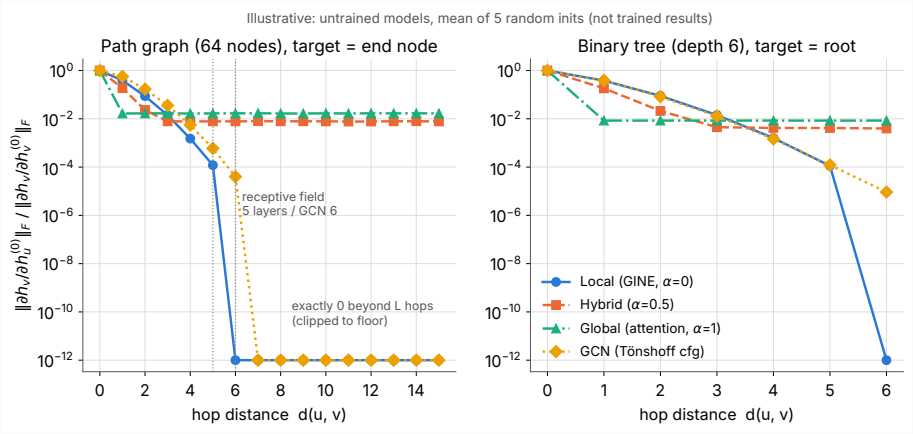
Pilot Reddit: virtual node vs SAGE by hop band [pilot]

Hop	n	SAGE	VMN	Cut
1	1,454	16,808	14,808	12%
2	17,876	90,056	56,115	38%
3	7,269	194,509	95,359	51%
4	613	207,113	77,806	62%
$\geq$5	57	326,334	62,993	81%

Mean test MSE over 100 seeds (raw units). Distance split by BFS from an anchor.

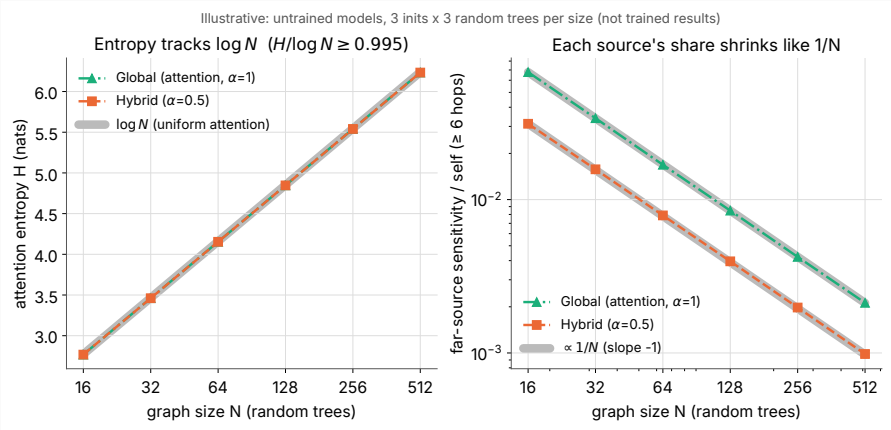
- The gain from a global shortcut (virtual node) grows with distance
- Caveats
  - Only VMN had an input projection (unfair baseline)
  - The target (character count) is almost a function of the node's own features
  - "Far" is confounded with "out of distribution"

Demo 1: Jacobian decays with hops [illustrative, untrained]



MP loses ≈3–10× per hop and is exactly 0 beyond L hops. Attention is flat at $\approx 1/N$. The $\alpha = 0.5$ hybrid follows the local curve for 2–3 hops, then settles on a floor half as high.

Demo 2: attention dilution with graph size [illustrative, untrained]



Entropy lies on $\log N$ ($H/\log N \geq 0.995$). Sensitivity to sources $\geq$6 hops away falls exactly as $1/N$, e.g. global: 6.8×10^{-2} at $N = 16$ and 2.1×10^{-3} at $N = 512$.

Demo 3: range-measure sanity on a 150-node path [illustrative, untrained]

Model (untrained)	Node-level range, hops (mean / max)	Graph-level Hessian range
Local $\alpha = 0$ (5 layers)	0.67 / 0.69 (≤ 5 ✓)	undefined (ReLU-only)
Hybrid $\alpha = 0.5$	18.7 / 25.8	not run
Global $\alpha = 1$	34.3 / 47.0	not run
GCN (6 layers, GELU)	0.89 / 0.89	17.7

- The node-level range respects the receptive field and grows once attention is mixed in
- The Hessian range reaches ≈18 hops, far past GCN's 6-hop reach: pooling plus a nonlinear head couples all pairs

$\Rightarrow$ Models are compared using the **node-level** range

Synthetic tree sweep v1: MP fails the 4-hop query

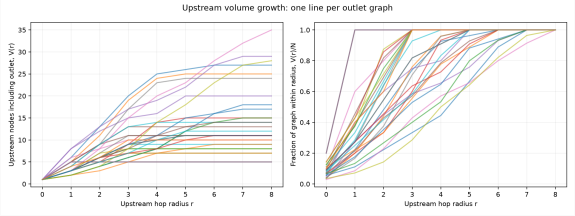
Model	Near	Far	Balanced
MP ($\alpha = 0$)	0.0036	1.0094	0.5065
$\alpha = 0.25$	**0.0024**	0.0129	**0.0077**
$\alpha = 0.50$	0.0074	0.0147	0.0110
$\alpha = 0.75$	0.0050	**0.0122**	0.0086
Attn ($\alpha = 1$)	0.0093	0.0189	0.0141

Test MSE; predicting zero gives 1. Trees of 63/127 nodes; 4 marked nodes at 1 hop, 4 at 4 hops; the root query picks near or far.

- MP solves the 1-hop query but fails the 4-hop one, though 4 hops lie inside its 8-layer reach
- Every hybrid solves it, and beats pure attention
- Confounded: different target draws, mean aggregation for a sum target, attention-only hop bias, unequal parameter counts

River experiment: setup

- LamaH-CE: 32 upstream gauge trees, 5–35 nodes (Klingler et al. 2021)
- Predict discharge 1/3/7 days ahead; inputs: weather, or weather + flow
- GRU encoder (90 days) $\rightarrow$ 8 spatial layers: $(1-\alpha)\,\mathrm{LN}(\mathrm{MP}) + \alpha\,\mathrm{LN}(\mathrm{Attn})$
- Baselines: outlet-only GRU, upstream pooling, persistence
- 72 fits, 3 seeds, $\approx$28.5 GPU-h
- Hypothesis: faster upstream growth $V(r)$ means a larger gain over MP



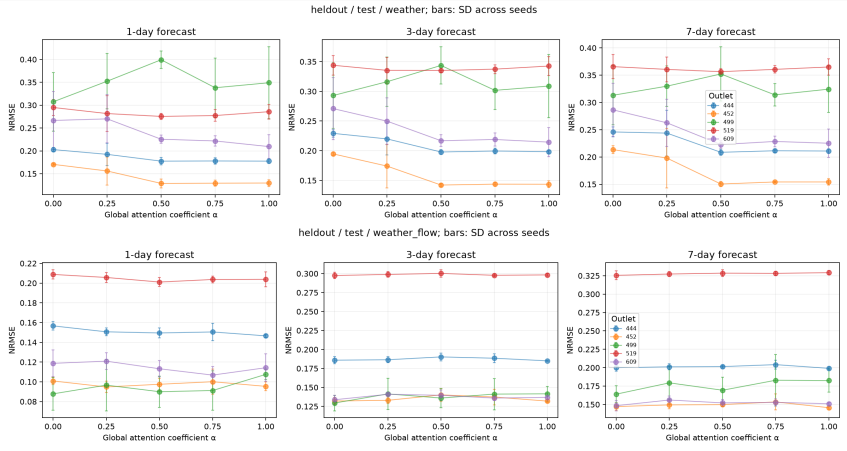
$V(r)$: upstream nodes within r hops. It saturates at N.

River held-out results: the α curve (test NRMSE, lower is better)

Model	Weather	Weather+flow
MP ($\alpha = 0$)	0.2666	0.1691
$\alpha = 0.25$	0.2628	0.1722
$\alpha = 0.5$	0.2489	0.1706
$\alpha = 0.75$	**0.2409**	0.1716
Attention ($\alpha = 1$)	0.2426	0.1713
Upstream pooling	0.2437	**0.1689**
Outlet-only GRU	0.2600	0.1696

- **Weather only**: $\alpha = 0.75$ is best, 9.6% better than MP
- But only 0.7% better than attention and 1.1% better than pooling: the gain comes mostly from *global access*
- **Weather+flow**: no gain; every $\alpha > 0$ is slightly worse than MP, pooling is best
- Mean NSE in the weather regime is negative (-0.30 at $\alpha = 0.75$)

River α curves per held-out outlet



Top: weather (most outlets improve up to $\alpha \geq 0.5$; the largest graph, 499, does not). Bottom: weather+flow (flat).

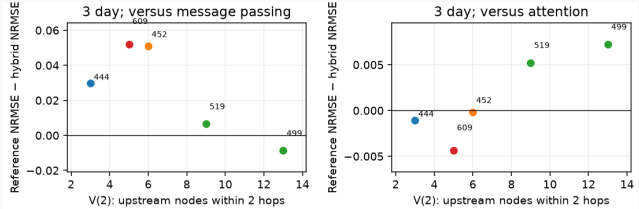
River context ablation: a hint of over-squashing

Weather	Full	Radius 2	Radius 0
MP ($\alpha = 0$)	0.2666	0.2673	**0.2584**
$\alpha = 0.75$	**0.2409**	0.2695	0.2741
Attention	**0.2426**	0.2583	0.2791

Held-out test NRMSE, mean over horizons. Radius r: dynamic inputs kept only within r hops.

- MP does *better* when upstream inputs are hidden
- Attention and the hybrid use upstream context: NRMSE about 13% and 12% lower than at radius 0
- Consistent with over-squashing, but also with optimisation or parameter effects
- The ablation masks inputs; it does not rewire the graph

River growth → gain hypothesis: not supported



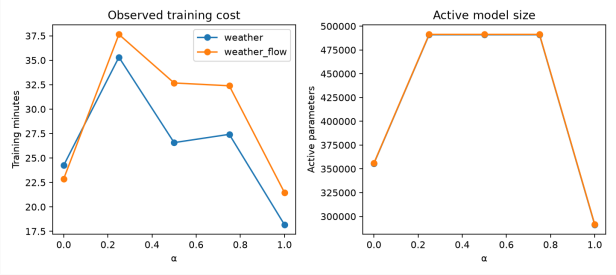
Held-out, weather, $\alpha = 0.75$, 3-day horizon. Positive = hybrid better than the reference.

Outlet	Nodes	$V(2)$	Gain vs MP
444	9	3	+13.1%
609	7	5	+18.8%
452	8	6	+26.2%
519	11	9	+2.9%
499	29	13	−4.4%

Gain averaged over seeds and horizons

- The fastest-growing outlet (499) gains *least*
- Spearman $\rho(V(2), \text{gain})$ is -0.70 vs MP (1- and 3-day) but $+0.90$ vs attention: **the sign flips with the reference**
- $n = 5$; leave-one-group-out ranges span both signs (-1.0 to $+0.5$)
- 27 development outlets, adjusted for size, depth and area: $\rho \approx 0$ for weather vs MP
- A graph-conditioned α selector is 2.3% *worse* than a fixed α

River cost vs α



Model	Active params	s/epoch
MP	355,715	52.9
Hybrid	490,979	80.1–80.6
Attention	291,299	54.2
Pooling	26,627	15.0
Outlet GRU	22,403	10.7

Training time only, RTX 3080 Laptop GPU

- A hybrid takes $\approx 1.5\times$ the time per epoch of either endpoint
- Pooling matches the best hybrid with $\approx 5\%$ of the parameters